Unit: Unit 0: Foundations of Algebra
Topic: Natural Numbers Integers

Name: ____________________
Date: _____________

Homework (Carolina Reaper)

Q1. A rhythm pattern repeats every 12 beats. If it starts on beat 0, on which beats will it repeat in the first 36 beats?
(A) $12, 24, 36$
(B) $0, 12, 24$
(C) $12, 24, 30$
(D) $0, 6, 12$
(E) $24, 36, 48$

Q2. Classify the number -15 as a:
(A) natural number
(B) whole number
(C) integer
(D) rational number
(E) irrational number

Q3. A painter uses a 5-note rhythm to create a pattern on a 25-note canvas. How many times will the pattern repeat?
(A) 5 times
(B) 6 times
(C) 4 times
(D) 3 times
(E) 1 time

Q4. Determine if -7 is an even or odd integer:
(A) even
(B) odd
(C) both
(D) neither
(E) cannot be determined

Q5. A musician plays a melody with a frequency of 200 Hz. If the frequency doubles every 2 seconds, what is the frequency after 6 seconds?
(A) 400 Hz
(B) 800 Hz
(C) 1600 Hz
(D) 6400 Hz
(E) 3200 Hz

Q6. What is the value of x in the equation $2x + 5 = 11$?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6

Q7. A composer writes a piece with a 12-beat rhythm. If the rhythm starts on beat 4, which beats will the rhythm repeat on in the first 36 beats?
(A) $16, 28, 40$
(B) $4, 16, 28$
(C) $12, 24, 36$
(D) $16, 28, 32$
(E) $4, 12, 20$

Q8. What is the next number in the pattern $2, 5, 8, 11, 14$?
(A) 15
(B) 17
(C) 18
(D) 20
(E) 21

Open Ended Questions (FRQ)

Q9. A music producer creates a 30-second audio clip with a tempo of 120 beats per minute. If the producer wants to add a 5-second drum solo every 10 beats, how many drum solos can fit in the audio clip? Show your work and explain your reasoning.

Q10. An artist creates a painting with a 20×15 grid. If the artist wants to divide the painting into smaller squares with areas of 5 square units, how many squares can fit in the painting? What is the perimeter of each square? Show your work and explain your reasoning.

Chef's Tips

- Encourage students to use music and art to visualize mathematical concepts
- Emphasize the importance of pattern recognition in mathematics and music
- Remind students to show their work and explain their reasoning for each problem